16.2 (Singularities and Zeros)

Types of Singularities :-

·) **Pole of order n :-**

Singularity is called a pole of order 'n' if Principal part of series has n terms (finite number of terms) with **negative powers. Then zo is the pole of order n.**

⇒ **Simple Pole :-** Pole of order "1" ✓

·) **Isolated Singularity :-**
is a singularity for which there exists a (small) real number such that there are no other singularities within a neighborhood.

In Short ⇒ There is no singularity in the neighborhood.

•) Essential Singularity :-

If Principal part has infinite number of terms.

Example :-

•) $e^{1/z}$

Expansion

$$= 1 + \frac{1}{z} + \frac{1}{z^2 2!} + \frac{1}{z^3 3!} + \cdots$$

→ $z = 0$ is the singular point

The principal part contain infinite terms and this above series has singular points in the neighbourhood. So,

Type of singularity

→ Essential singularity

→ Isolated singularity

•) $f(z) = \frac{1}{z(z-2)^5} + \frac{3}{(z-2)^7}$

Singular points :-

•) $z = 0$

•) $z = 2$

Type of singularity :- Poles

•) $z = 0 \Rightarrow$ Simple Pole

•) $z - 2 \Rightarrow$ Pole of Order "5"

$\longleftrightarrow$

* **Poles of order :-** "highest degree of principal part"

Example 2:-

Behaviour Near a pole:-

$$f(z) = \frac{1}{z^2}$$

Singular Point:

$z = 0$

Type of Singularity:-

- Poles

$z = 0 \Rightarrow$ Pole of Order "1"
(**Simple Pole**)

Theorem 1

Poles:-

- If $f(z)$ is analytic and has a pole at $z = z_0$, **then**

$$|f(z)| \rightarrow \infty \text{ as } z \rightarrow z_0 \text{ in any manner}$$

Q) $z^2+4=0$

$\rightarrow z = 2i, -2i$ are two zeros (where function zero)

Zero of Analytic functions:-

- A zero of an analytic function $f(z)$ in a domain D is a $z = z_0$ in D **such as / that $f(z_0)=0$.**

- A zero has order **n** if not only **f but also the derivatives** $f', f'', ----, f^{(n-1)}$ are all zero at **$z = z_0$ but $f^{(n)}(z_0) \neq 0$.**

- **Simple Zero:-**
A first-order zero is called Simple Zero.
when $f(z_0)=0$

$\rightarrow$ In other words, function's derivative is not zero

- **Second Order Zero:-**
$\rightarrow$ for second order, $f(z_0) = f'(z_0) = 0$ but $f''(z_0) \neq 0$

Q1) $\sin^4 \frac{1}{2} \times z$

location and order of zeros:-

$f(z) = \sin^4 \frac{z}{2}$

location:-

where $f(z)$ is "0"
so location is $n\pi$ where
n is any integer.

Order of zero:-

•) $f'(z) = 4\sin^3 \frac{z}{2} \times \cos\frac{z}{2} \times \frac{1}{2}$

$f'(z) = 2\sin^3 \frac{z}{2} \cdot \cos\frac{z}{2}$

$f'(n\pi) = 0$

•) $f''(z) = -2\sin^3 \frac{z}{2} \sin\frac{z}{2} + 2\cos\frac{z}{2} \times 3\sin^2\frac{z}{2} \cos\frac{z}{2} \times \frac{1}{2}$

$f''(z) = -2\sin^4 \frac{z}{2} + 3\cos^2\frac{z}{2} \sin^2\frac{z}{2}$

$f''(n\pi) = 0$

Similarly,

- $f'''(\cancel{\theta}\ n\pi) = 0$
- $f^{4}(\cancel{\theta}\ n\pi) \neq 0$

→ So, order of zero is "4"

Removable Singularity :-

o) A removable singularity is a singular point of a function for which it is possible to assign a complex number in such a way that becomes analytic

o) We say that function $f(z)$ has a removable singularity at $z = z_0$ if $f(z)$ is not analytic at $z = z_0$, but can be analytic there by assigning a suitable value $f(z_0)$. Such singularities are of no interest since they can be removed.

Example :-

$$\frac{\sin z}{z}$$

$$\frac{\sin z}{z} = \frac{1}{z}\left[z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots \right]$$

Scanned with CamScanner

*

$$\frac{\sin}{z} = \left\{1 - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots\right]$$

↳ So this is removable singularity

We can make it analytic by assigning the function its limiting value.

$$\lim_{z \to 0} \frac{\sin z}{z} = 1$$

↳ (making it analytic at $z=0$)

Scanned with CamScanner

Theorem 1 (Poles)
Example :-

$$f(z) = \frac{1}{z^2}$$

Singular Point :

$$\boxed{z = 0}$$

Type :-
- Poles of order "2"

"Substitution"

$$\boxed{f(z) = \frac{1}{w}}$$

$$f(z) = \frac{1}{z^2}$$

Using $f\left(z = \frac{1}{w}\right)$

so,

$$f(1/w) = \frac{1}{1/w^2}$$

$$f(1/w) = w^2$$

behaviour of w at "0"

This $f(1/\omega)$ is analytic at "0", (~~Thus~~ So, $f(z)$ is analytic at ∞)

Scanned with CamScanner

18/Oct/22

Determine the location of singularities at including those at infinity.

13-22

Q13)

$$\frac{1}{(z+2i)^2} - \frac{z}{z-i} + \frac{z+1}{(z-i)^2}$$

Singularity at:

$z = -2i$. Pole of order 2

$z = i$ Pole of order 2

Singularity at infinity:

$$z = \frac{1}{w}$$

$z \to \infty$

$w \to 0$

So

$$= \frac{1}{\left(\frac{1}{z}+2i\right)} - \frac{z}{z-i} + \frac{z+1}{(z-i)^2}$$

Scanned with CamScanner

$$= \frac{1}{\left(\frac{1}{w}+2i\right)^2} - \frac{\frac{1}{w}}{\left(\frac{1}{w}-i\right)} + \frac{\frac{1}{w}+1}{\left(\frac{1}{w}-i\right)^2}$$

$$= \frac{1}{\left(\frac{1}{w}+2iw\right)^2} - \frac{\frac{1}{\cancel{w}}}{\frac{1-iw}{\cancel{w}}} + \frac{\frac{1+w}{\cancel{w}}}{\frac{(1-iw)^2}{w^2}}$$

$$= \frac{w^2}{(2+iw)^2} - \frac{1}{1-iw} + \frac{1+w}{\frac{(1-iw)^2}{w}}.$$

$$= \frac{w^2}{(2+iw)^2} - \frac{1}{1-iw} + \frac{(1+w)\times w}{(1-iw)^2}$$

at $\boxed{w=0}$

$$= 0 - 1 + 0$$

$$= -1$$

→ There is no singularity at $w=0$ so the given function has no singularities at infinity.

$\cdot)\ \cot^4 z$

$$\left(\frac{\cos z}{\sin z}\right)^4$$

Singularity :

where $\sin z$ is "0"

so $n\pi$ is singularity $n \in Z$

Singularity at infinity :-

$$z = \frac{1}{w}$$

$z \to \infty$

$w \to 0$

$\cdot)\ \cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots$

$\cdot)\ \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!}$

So,

$$\cos\left(\frac{1}{w}\right) \doteq 1 - \frac{1}{w^2}\frac{1}{2!} + \frac{1}{w^4}\frac{1}{4!} - \frac{1}{w^6}\frac{1}{6!}$$

$$+ - - -$$

and

$$\sin\left(\frac{1}{w}\right) = \frac{1}{w} - \frac{1}{w^3}\frac{1}{3!} + \frac{1}{w^5}\frac{1}{5!}$$

$$- \frac{1}{w^7}\frac{1}{7!} + - - - -$$

The Behaviour at $\boxed{w=0}$

So, at $w=0$ the $\left(\frac{\cos z}{\sin z}\right)^4$ has singularity which is essential because of infinite terms of Principal part

Result :-

So, the function has essential singularity at infinity.

·) e^{1/z^2}

$$= 1 + \frac{1}{z^2 2!} + \frac{1}{z^3 3!} + \cdots$$

→ at $z = 0$

// essential singularity //